Communication and Multimedia

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1. The Communication Process

1.1. Two Main Actors: Sender and Receiver

The communication process is built around two primary actors: the **Sender** and the **Receiver**.

- The **Sender** has an idea or intention that they want to share.
- This idea must be encoded into a **message** that can be transmitted.
- The **Receiver** receives the message and interprets it, transforming it into **information**.

The information reconstructed by the receiver can never correspond **100%** to the sender's original idea. Interpretation is always influenced by the receiver's knowledge, context, and perception.

Example: A teacher explains a concept in class (sender → message). Each student (receiver) understands it slightly differently depending on their background knowledge.

1.2. Message and Physical Support

For a message to travel from sender to receiver, it needs a **physical support**. This support is the material carrier of the message. It may be **a screen** (for digital text or video), **a paper** (for printed communication) pr a **a haptic vibrations** on a device (such as a game controller).

It ensures that the abstract idea takes on a tangible form that can be perceived by human senses.

Example: A love letter written on paper versus the same words sent in a text message—different supports, but the same underlying intent.

1.3. Forms of the Message

The **content of the message** can take various forms:

- **Text** (a written paragraph, an email)
- **Speech** (a spoken explanation, a voicemail)
- **Image** (a photograph, a diagram)

Each form is considered a **channel** when it is uniform, homogeneous, and continuous.

Example*:* A news broadcast may combine multiple channels: a journalist's voice (audio channel), text displayed on the screen (text channel), and supporting images (visual channel).

1.4. Channels and Human Perceptual Dimensions

Every **channel** belongs to a broader **dimension**, which corresponds to a specific human perceptual domain:

- **Vision** → text, images, videos.
- **Hearing** → speech, music, sound effects.
- **Touch (Haptics)** → vibrations, tactile feedback.

Combining different channels across these dimensions makes the communication richer and more engaging.

Example*:* In virtual reality, a game may use multiple dimensions: visual (3D graphics), auditory (sound effects, voice), and haptic (controller vibrations).

☑ In summary:
→ Communication begins with a **sender's idea**, encoded into a **message**.
→ The message travels via a **support**, in a particular **form**.
→ The form defines the **channel**, which belongs to a **human perceptual dimension**.
→ The **receiver interprets** the message to build **information.**
→ The information will never be an exact replica of the sender's idea.

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2. Media vs Multimedia

2.1. Media, plurial of Medium (sense 1: means of communication)

Technical means used to transmit a message or information from a sender to a receiver.
Example: speech, writing, an email.

2.2. Media (sense 2: mass media)

Refers to **organized systems of mass communication**, designed to reach a wide audience.
Example: newspapers, radio, television, the Internet.

2.3. Multimedia

Multimedia refers to the **combined use of several media or forms of communication** (text, image, sound, video, animation, interactivity).
Example: an educational website containing text, explanatory videos, interactive images, and audio.

2.4. Multimedia application

A digital application that uses several media to communicate with the user.
A **medium** corresponds to a **channel**: a unique flow defined within a single perceptual dimension (such as sound, vision, or touch).

3. Perception and perceptual Dimension

Perception refers to the **sensory experience of the internal world** (self-perception) and the **external world** (perception of others).
It involves the five senses (touch, sight, hearing, smell, taste), as well as **proprioception** (the detection of body movements) and **bodily and emotional feeling** (such as sensing cold, fear, or hunger).
It enables the identification and interpretation of both internal and external stimuli through cognitive functions such as memory.
A **perceptual dimension** is defined as a unique and consistent stream of sensory input within the overall perceptual process.

4. Hypermedia and Non-linear Navigation

Hypermedia is a system that connects multiple media through links, enabling non-linear navigation within a multimedia application.
Non-linear navigation is a **non-sequential path** through a multimedia application, made possible by **links**.

4.1. Benefits of Non-Linear Navigation

- **User Control and Freedom**
  - Users can choose their own path through the content.
  - **Example**: On an educational website, learners can jump directly to the topic they find most relevant.
- **Personalized Learning/Experience**
  - Each user can adapt the navigation to their needs, interests, or pace.
  - **Example**: In an e-learning platform, a student can skip basics and focus on advanced modules.
- **Exploration and Engagement**
  - Encourages curiosity and active exploration rather than passive consumption.
  - **Example**: Interactive museum kiosks where visitors explore artifacts via images, videos, and maps.
- **Efficient Access to Information**
  - Users can reach specific information quickly without going through unnecessary steps.
  - **Example**: Hyperlinked FAQs let you jump directly to the answer.

4.2. Drawbacks of Non-Linear Navigation

- **Risk of Disorientation**

  - Too many links or pathways may confuse the user.
  - **Example**: A website with excessive cross-links where users lose track of where they started.
- **Incomplete Understanding**
  - Skipping content may cause users to miss important steps or background knowledge.
  - **Example**: In an online course, learners may jump to advanced topics without mastering the basics.
- **Cognitive Overload**
  - Too much choice and too many navigation options can overwhelm the user.
  - **Example**: A multimedia application with multiple menus, pop-ups, and links can distract from the main message.
- **Design Complexity**
  - Developing a non-linear structure is harder for designers and requires careful planning.
  - **Example**: Ensuring all hyperlinks are relevant, functional, and don't create "dead ends."

In Summary
Benefits: Freedom, personalization, engagement, efficiency.
Drawbacks: Disorientation, incomplete learning, overload, design complexity.
The key is balance: combining non-linear navigation with some structured guidance (like menus, maps, or recommended paths) to support both freedom and clarity.

5. Multimedia application architecture

Every multimedia application can be understood as a **simulation**, whether it represents a physical phenomenon or object, or a mental abstraction. In all cases, it is a **digital representation** of an existence, either concrete or abstract.
The process of creating a digital version of an idea, a phenomenon, or an object is called **computer simulation**. It refers to the transition from physical reality (defined as a reality that requires no technological device to be perceived) to **virtual reality** (defined as a reality that requires interfaces to be perceived).

5.1. The scene

A multimedia application is composed of one or more **scenes**. Each scene is defined through **perceptual dimensions** and **hypermedia links** that enable navigation. The data of a scene are described by **metadata**, which specify both the format and the nature of the information. Finally, these data are interconnected by **semantic links**, ensuring the fluidity and coherence of the scene.
A scene can be defined as a **temporal decomposition** of multimedia content, defined by the discontinuities within the overall content flow.

5.2. Semantic coherence

In the context of **multimedia applications**, **semantic coherence** means that all the different media elements (text, image, video, sound, animation, haptics, etc.) work together to convey **a consistent meaning** to the user.
Semantic coherence is the degree to which the various components of a multimedia application remain logically and meaningfully aligned with the intended message, scenario, or concept.

- In practice
  - The **text** on a slide should match the **images** shown, rather than contradicting or confusing the user.
  - **Background music** should reinforce the mood of the scene (e.g., calm for learning, dynamic for action), not distract from it.
  - **Animations** should illustrate the same concept that the narration is explaining.
- **Why it matters**

If semantic coherence is broken (for example: cheerful music in a serious medical training simulation, or images that contradict the spoken explanation), the user experiences **cognitive dissonance** or **confusion**, reducing comprehension and learning efficiency.